medsum-fact — Review 2 Baseline Report

Plain 4-bit QLoRA fine-tune of Llama-3.1-8B-Instruct for dialogue → clinical note

Run **baseline_qlora_llama31_8b** · mode **full** · Kaggle kernels `hirthick17/medsum-baseline-qlora` (train) → `hirthick17/medsum-baseline-infer` (infer) · generated 2026-09-21T21:36:23 from repo commit a7504b8

1. Run summary

The baseline defined in REVIEW2_BASELINE.md: no fact table, no ORPO, no verifier, no provenance. One instruction-tuned decoder, 4-bit NF4 QLoRA SFT on the assistant tokens only, greedy inference on the six official test splits.

Field	Value
Base model	unsloth/Meta-Llama-3.1-8B-Instruct ungated mirror of meta-llama/Llama-3.1-8B-Instruct (Llama 3.1 Community Licence)
Quantization	4-bit NF4, double-quant=True, compute torch.float16
LoRA	r=16, α=32, dropout=0.05, targets: q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj trainable 41.9 M of 8.07 B (0.52 %)
Data	train 4265 · val 220 · max_seq_len 4096 mix: {"dialogue_soap": 3000, "mts_dialog": 1198, "aci_bench": 67} · leakage check: passed
Optimisation	lr 0.0002 · cosine · warmup 0.03 · paged_adamw_8bit · batch 1 × accum 16 · 2 epochs · seed 42 · liger fused-CE=True
Hardware (train)	2x Tesla T4 · 29.1 GB total · device_map balanced · peak — GB / GPU · wall time **— h**
Hardware (infer)	2x Tesla T4 · peak 10.49 GB / GPU · wall time **5.37 h**
Versions	torch 2.10.0+cu128 · transformers 4.57.6 · peft 0.19.1 · bitsandbytes 0.50.2 · python 3.12.13

2. Training

Metric	Value
Steps completed	332 (max_steps=534, epoch 1.24)
Final train loss	0.7403
Best eval loss	0.9555 at step 200
Eval loss (first → last)	1.1180 → 0.9614
Wall-clock guard	7.5 h — HIT (stopped early: 332 of 534 planned steps)

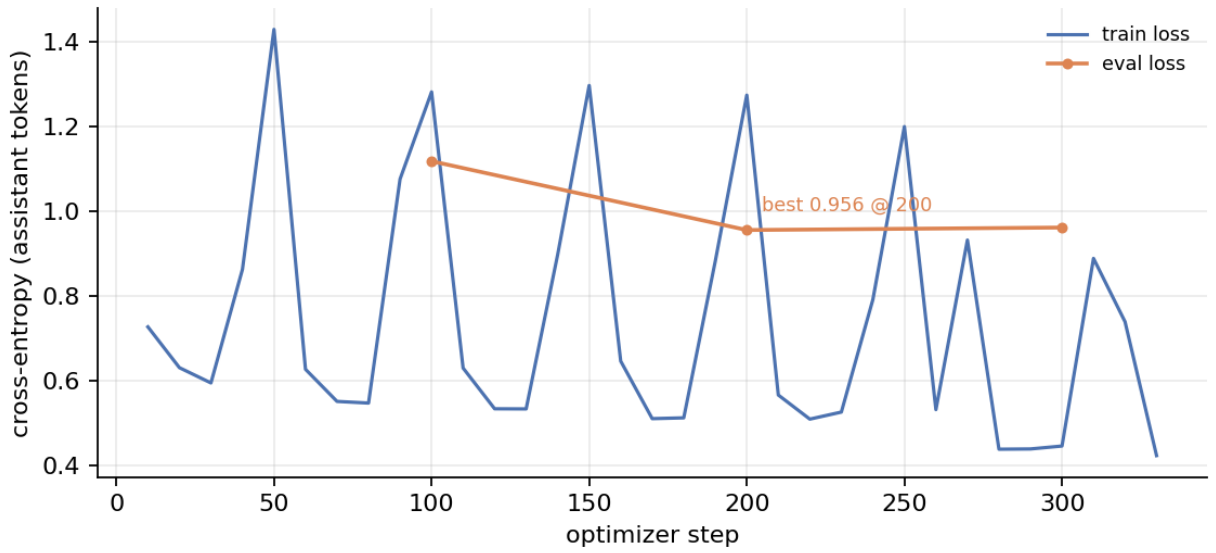


Figure 1 — assistant-token cross-entropy on train (every logging step) and on the 220-item validation set (every eval step). load_best_model_at_end selects the checkpoint at the lowest eval loss for generation.

3. Validation checks (REVIEW2_BASELINE.md §7)

Check	Result	Expectation
Loss report	final train 0.740 · best eval 0.956 @ 200	eval loss decreasing, best step recorded
Length ratio (pred / ref words)	1.290	0.8 – 1.2
Quick ROUGE-L on val subset (n=20)	14.7	sanity gate only
Truncated / empty on val	9 / 0	0 / 0
Adapter reload test	fresh kernel · adapter = checkpoint not mounted · 0 empty outputs	generation from a fresh PeftModel.from_pretrained, not the training process
Zero-shot base control (§7 step 6, same 20 items)	ROUGE-L 21.2 vs adapter 14.7 · len ratio 2.781 vs 1.290 · truncated 0 vs 9	adapter should be more note-like: higher ROUGE-L, ratio nearer 1
ROUGE-L gate · aci_bench val	10.5 — BELOW GATE	≥ 35
ROUGE-L gate · mts_dialog val	18.8 — BELOW GATE	≥ 30

Reload note: generation runs in a separate kernel from training (fresh base + PeftModel.from_pretrained), which is the §7 step 5 reload test itself; non-quantised params upcast to fp32 as in training

4. Test-set generation (REVIEW2_BASELINE.md §8)

Greedy decoding (num_beams=1, do_sample=False, repetition_penalty=1.2, max_new_tokens=1024, 6-gram repeat block on generated tokens). ROUGE F1 × 100 with Porter stemming, computed here as a first look — Person 3's evaluation (BERTScore, coverage, hallucination, negation, numeric consistency) supersedes these. DialogueSOAP is synthetic and is not a headline number.

Split	n	R-1	R-2	R-L	len ratio	gen tok	trunc	empty	status
aci_bench_test_taskB_1	40	21.9	6.6	11.4	1.26	895	29	0	headline
aci_bench_test_taskC_2	40	20.9	6.2	10.9	1.09	916	28	0	headline
aci_bench_test_taskC_3	40	23.0	7.2	12.3	1.00	887	27	0	headline
mts_dialog_test_chat_2023	200	34.8	13.7	29.7	1.85	78	5	0	headline
mts_dialog_test_sum_2023	200	34.1	11.3	28.6	1.36	70	6	0	headline
dialogue_soap_test	250	46.7	16.4	29.7	0.97	353	7	0	synthetic
headline (weighted)	520	31.6	11.1	25.1	1.49	264	95	0	

5. Handoff checklist (REVIEW2_BASELINE.md §10)

	Item	Detail
☑	adapter used for generation == final/adapter_model.safetensors (sha256)	used 14e712617f1456b6 | final/ 14e712617f1456b6
☑	predictions/baseline_qlora/: 6 files, 770 lines, 0 JSON errors	6 files, 770 lines, 0 JSON errors
☑	truncated == true count per split (reported for Person 3 to flag/exclude)	29 / 28 / 27 / 5 / 6 / 7
☑	training_config.json matches trainer_state.json (steps, best_step)	config steps=332 best=200 | state global_step=332
☑	train_manifest sha256 matches kaggle/dataset/baseline_{train,val}.jsonl	verified
☑	final/ adapter reloaded in a fresh kernel and generated on-task output (§7 step 5)	sha256 14e712617f1456b6 = checkpoint not mounted | matches train_summary: None | empty outputs 0
☑	no test id appears in baseline_train/val.jsonl	0 overlapping ids
☑	environment.txt present	pip freeze captured
☑	final/ adapter present	160.1 MB

☑ verified · ☒ failed · ☐ could not be verified from this machine

6. Sample predictions

One item per split, chosen as the record whose ROUGE-L is closest to that split's mean (representative, not cherry-picked). Text is clipped to 900 characters.

aci_bench_test_taskB_1 — aci_bench_test_taskB_1_D2N099-virtscribe · prompt 2403 tok · generated 898 tok

Reference	Prediction
CHIEF COMPLAINT Cough, nasal congestion, and rhinorrhea. HISTORY OF PRESENT ILLNESS Randy Gutierrez is a 9-month-old male who is being evaluated today for a cough, nasal congestion, and rhinorrhea. He is accompanied by his mother. Symptoms began yesterday around dinnertime with clear nasal drainage, pulling on his ears bilaterally, and rubbing his nose frequently. He does not have a barky cough or a sore throat. His rectal temperature was 100.3 degrees. The patient did consume a 7 ounce bottle this morning, but mostly denied the next offered bottle and declined cereal. He is irritable and having trouble falling asleep, described “like he is fighting it”. The patient has been exposed to an older sister who is sick and a cousin who has a sinus infection. Treatment has included saline nasal spray, Tylenol, baby Vick's, and a humidifier. When the patient aroused this morning, he was wa …	CHIEF COMPLAINT: The chief complaint is Randy Gutiérrez, DOB 08/10/20, presents with URI. HISTORY OF PRESENT ILLNESS: Randy Gutiérrez, D.O.B. 08/10/2020, presented to clinic today complaining of having a cold. He states that he began to develop these symptoms approximately yesterday evening during dinner. The mother reports that she first noted nasal congestion followed by rhinorrhea. She denies purulent discharge. Denies ear pain. Mother notes that he has had irritability and difficulty sleeping. No history of recent travel or contact with someone ill. His vital signs were stable. Physical examination revealed bilateral conjunctival injection bilaterally. Examination of the tympanic membranes showed normal landmarks without erythema or bulging. Nasopharynx shows redness and swelling. Oral cavity shows mild erythematous tonsils. Neck supple. Respiratory system reveals decreased air en …

aci_bench_test_taskC_2 — aci_bench_test_taskC_2_D2N136-virtassist · prompt 1319 tok · generated 1024 tok · TRUNCATED

Reference	Prediction
CHIEF COMPLAINT Joint pain. HISTORY OF PRESENT ILLNESS Ms. Janet Sullivan is a 33-year-old female who presents to the clinic today for evaluation of joint pain. She localizes the joint pain to her bilateral knees. The patient reports she has been doing stair workouts, which she thought was building muscle on her knees. She denies any pins and needles in her feet or swelling in her legs. The patient notes that she would look out for those symptoms because she has diabetes. She denies any fever or chills. She denies any nausea or vomiting. The patient denies any chest pain or shortness of breath. She notes she had a right shoulder rotator cuff repair about 2 years ago, but she has not had any problems since that point. She reports that her diabetes are well-controlled. The patient does have high blood pressure but she monitors it and takes Norvasc 5 mg a day. She notes her blood pre …	CHIEF COMPLAINT: The patient is a 33 y/o F who comes in complaining of bilateral knee pain. HISTORY OF PRESENT ILLNESS: Janet is a 33 y.o.f. who presents to clinic complaining of bilateral knee pain. She states she has been working out stairs lately trying to build her leg muscles. However, she denies any improvement in her condition. The patient also complains of bilateral knee pain bilaterally without any history of trauma. No fevers, chills, night sweats, weight loss, hemoptysis, orthopnea, PND, palpitations, dyspnea, pleuritic chest pain, abdominal pain, diarrhea, constipation, arthralgias, arthritis, rash, pruritus, wheezing, hoarseness, cough, sputum production, syncope, weakness, fatigue, tremors, numbness, tingling, paresthesia, diplopia, photophobia, headache, visual changes, hearing loss, vertigo, tinnitus, dysphagia, odynophagia, epigastric burning, regurgitation, nausea, v …

aci_bench_test_taskC_3 — aci_bench_test_taskC_3_D2N185-virtscribe · prompt 3242 tok · generated 961 tok

Reference	Prediction
CHIEF COMPLAINT Evaluation of abnormal renal ultrasound with atrophic right kidney. HISTORY OF PRESENT ILLNESS Next patient Ashley James is a 14-year-old female, previously healthy, who is here for evaluation of abnormal renal ultrasound with atrophic right kidney. The father details a history of the patient coming home sick one day with extreme right-sided abdominal pain. Her parents brought her for evaluation, and they treated her for a stomach virus and administered fluids prior to sending her home. The next day, the patient's pain was more severe. The patient returned to the doctor's office and an ultrasound and additional testing was performed which revealed left kidney hypertrophy. At that time, they were encouraged to follow-up with her primary care physician. The patient was seen briefly by her physician the next day, which was insignificant. Ashley was still experiencing sto …	CHIEF COMPLAINT Abnormal Renal Ultrasound with Atrophy Right Kidney. HISTORY OF PRESENT ILLNESS The patient Ashley James is a 14-year-old female who comes in for further evaluation of abdominal pain and vomiting. She states that she has had intermittent episodes of nausea and vomiting for several weeks. The patient denies dysuria, frequency, urgency, hematuria, flank pain, or gross hematuria. No recent travel or contact with anyone ill. Denies tobacco, alcohol, illicit drug use. Family History positive for renal disease including mother, maternal grandmother, paternal uncle, and cousin. She has had previous evaluations by pediatrician and gastroenterologist without findings suggestive of appendicitis or inflammatory bowel disease. Her vital signs include BP 110/70 mm Hg; HR 80 bpm; RR 16/min; Temp 98.2°F (36.8°C). Physical examination shows generalized tenderness in abdomen, negative …

mts_dialog_test_chat_2023 — mts_dialog_test_chat_2023_47 · prompt 284 tok · generated 8 tok

Reference	Prediction
A 2-month-old female with 1-week history of congestion and fever x2 days.	Congestion with fever x2 d.

mts_dialog_test_sum_2023 — mts_dialog_test_sum_2023_62 · prompt 157 tok · generated 4 tok

Reference	Prediction
Intractable migraine with aura.	Classic migraine headache.

dialogue_soap_test — dialogue_soap_test_201 · prompt 452 tok · generated 235 tok

Reference	Prediction
S: The patient, a male, presents with a 2-day history of abdominal pain, decreased appetite, and vomiting. He reports the pain is localized to the stomach area. O: Physical examination revealed tenderness and pain on the right side of the abdomen. Laboratory tests confirmed a positive SARS-CoV-2 PCR test from a nasopharyngeal swab. No additional lab values or imaging results provided. A: The primary diagnosis is appendicitis, with a differential diagnosis of a gastrointestinal manifestation of COVID-19 due to the positive SARS-CoV-2 test. The presence of right-sided abdominal tenderness and systemic symptoms such as vomiting support the diagnosis of appendicitis. P: The patient will be started on empiric intravenous antimicrobial treatment with cefotaxime and metronidazole. A laparoscopic appendectomy is scheduled urgently to remove the inflamed appendix. Post-operative care will include …	S: The patient presented to the emergency department complaining of abdominal pain for the last two days, nausea, vomiting, and decreased appetite. She has not eaten much over the past couple of days due to her discomfort. O: Physical exam revealed right lower quadrant tenderness upon palpation. Laboratory findings were significant for COVID-19 positivity by PCR testing from a nasopharyngeal swab. No other abnormalities noted on vital signs or additional labs at this time. A: Primary assessment suggests acute appendicitis complicated by COVID-19 pneumonia given the recent history of feverish illness and respiratory symptoms. Differential diagnosis includes perforated viscus secondary to appendicitis versus non-perforating inflammatory process. Urgent surgical intervention indicated due to high suspicion of appendiceal rupture. P: Initiate IV antibiotics with cefotaxime and metronidazole …

7. Environment

accelerate 1.13.0 · bitsandbytes 0.50.2 · liger_kernel 0.8.3 · peft 0.19.1 · rouge_score 0.1.2 · torch 2.10.0+cu128 · transformers 4.57.6

Run directory: D:\OneDrive\Documents\B.Tech\3rd Year\AI\Projects\medsum-fact\outputs\kaggle\outputs\baseline_qlora_llama31_8b. Predictions and adapter stay under outputs/ (git-ignored; corpora licences forbid redistribution). This PDF, metrics.json, training_config.json and trainer_state.json are committed under reports/.